

Unit 10, Lesson 2: Creating Histograms

Benchmark

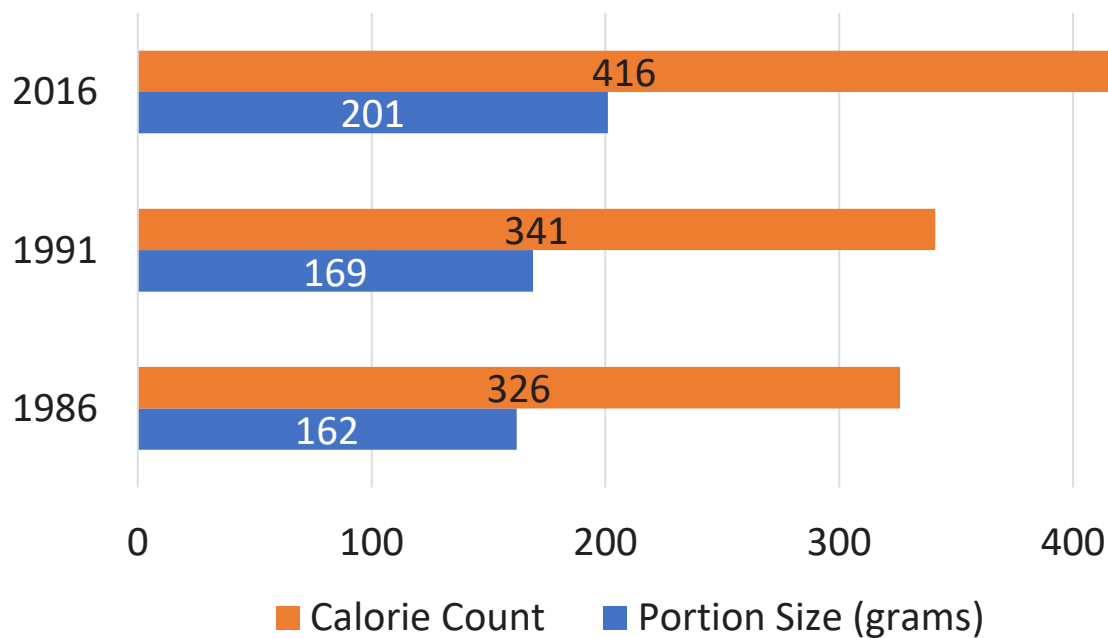
MA.6.DP.1.5 Create box plots and histograms to represent sets of numerical data within real-world contexts.

Learning Target

- ☐ I can create a histogram to represent data.

10.2.1 Warm-Up: Notice and Wonder

The portion size and calorie count of fast-food menu items have changed over time.



1. What are two things you notice about the graph?
2. What is one thing you wonder about the graph?

10.2.2 Guided Instruction: Elements of Creating Histograms

1. An earthworm farmer set up several containers of a certain species of earthworm so he could learn about their lengths. The farmer measured the lengths of 24 earthworms in one of the containers. Each length was measured in millimeters and recorded in the table.

6.5	11.2	18	19.6	20.1	23.8	23.5	25
25.7	26.2	27.5	27.5	28	29.9	32	33
41.4	42.3	48.8	52	54.4	59	60.6	77

Work with your partner to collect the pieces needed to create a histogram.

- a. The range of the data is _____.

Determine the **interval width** by dividing the range and the number of intervals desired.

$$\frac{\text{Range}}{\text{Number of Intervals}} \approx \text{Interval Width}$$

- b. Discuss three different numbers of intervals you could use with your partner, and how using a different number of intervals would affect the histogram.
- c. For this histogram, use six intervals. Divide the range by _____ and round up to the nearest whole number. This gives an **interval width** of _____.



A **frequency table** is a table that shows how often each item, number, or range of numbers occurs in a set of data.

Use the steps below to create a frequency table to represent the data.

- d. Complete each statement.

The minimum value is _____, so create the first interval that starts at 6 (or less).

To get the highest value in the interval, add the interval width of 12.

The first interval is 6 – 18, the second interval is 18 – _____.

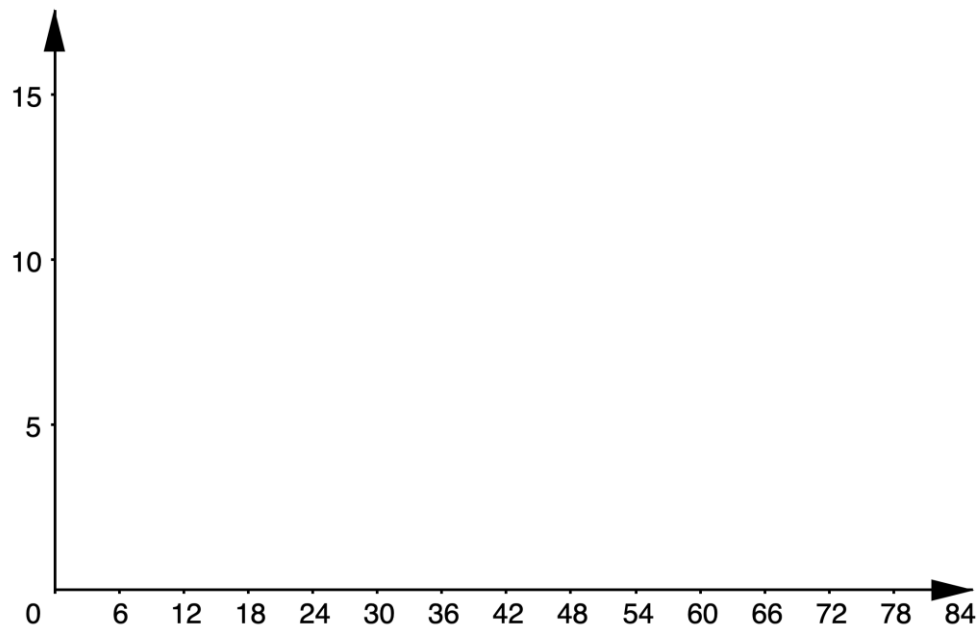
- e. With your partner, discuss if the value of 18 can be included in both intervals. Explain your conclusion.
- f. Complete the **Length (mm)** column of the table shown to indicate the intervals.

Length (mm)	Frequency
6 – 18	
18 –	

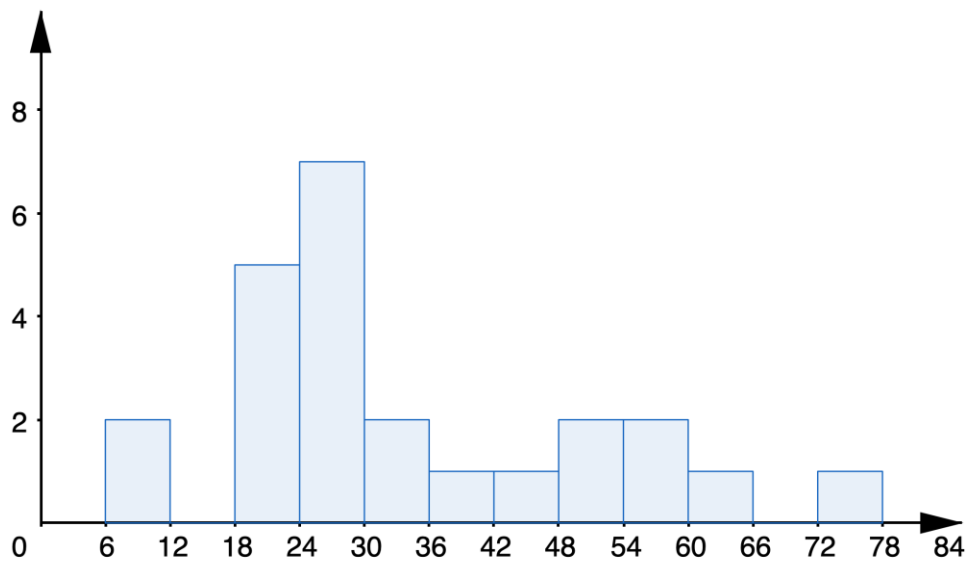
- g. With your partner, determine the frequency values and fill in the **Frequency** column of the table.

In a histogram, the bars touch. The variable measured is placed on the horizontal axis, and frequency is placed on the vertical axis.

- h. Label the horizontal and vertical axis of the grid below. Then, use the grid to create the histogram to display the lengths of earthworms in millimeters.



2. The same data could be displayed in this histogram.



- a. With your partner, discuss the differences you notice about this histogram from the one you created.
- b. Explain which histogram better displays the data.
- c. Ask two classmates the reasons they listed for which histogram better displays the data. Record their choice and write your summary of their reason. *You are the only person who should write in the first two columns of the table.* Have the person initial next to your summary stating that your summary was correct.

Histogram Choice	My Summary of their Reason	Initials

- d. Explain whether the data has the characteristics that will result in truthful data.

10.2.3 Guided Practice: Creating Histograms

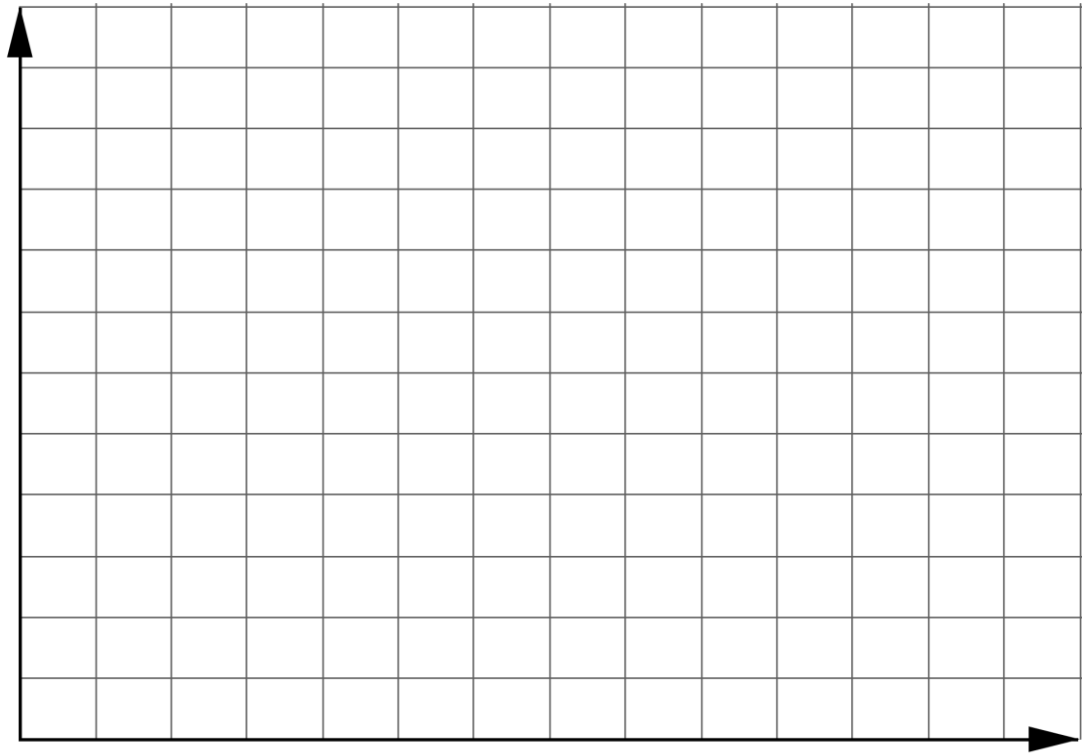
1. The data shows the number of Snapchats sent by 12 seventh-grade students in one day.

99, 97, 94, 88, 84, 81, 80, 77, 71, 63, 60, 54

Complete the following to create a frequency table and a histogram to represent the number of Snapchats sent by seventh-grade students in one day.

- a. How many intervals are appropriate to use for this histogram?
- b. List three possible interval widths.
- c. Using one of the possible interval widths, create the frequency table and histogram of the data. Make sure to label both the horizontal and vertical axis of the histogram.

Snapchats	Frequency



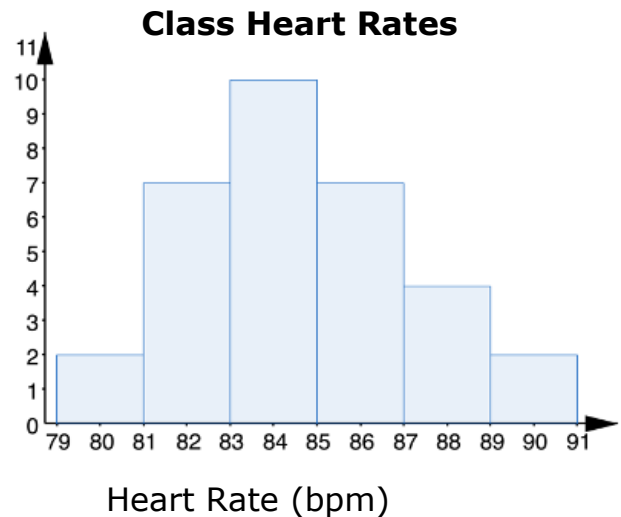
- d. Find a classmate who used a different interval width and compare your histograms. Summarize your discussion on what you notice about the two different graphs.

2. Raisa used data she collected about the resting heart rates, in beats per minute (bpm), of 22 classmates to create a histogram. The heart rates are listed below.

89 87 85 84 90 79 83 85 86 88 84 81 88 85 83
83 86 82 83 86 82 84

She created a frequency table and used it to make a histogram. The frequency table and histogram Raisa created are shown below. Unfortunately, Raisa noticed the data she collected was not accurately displayed.

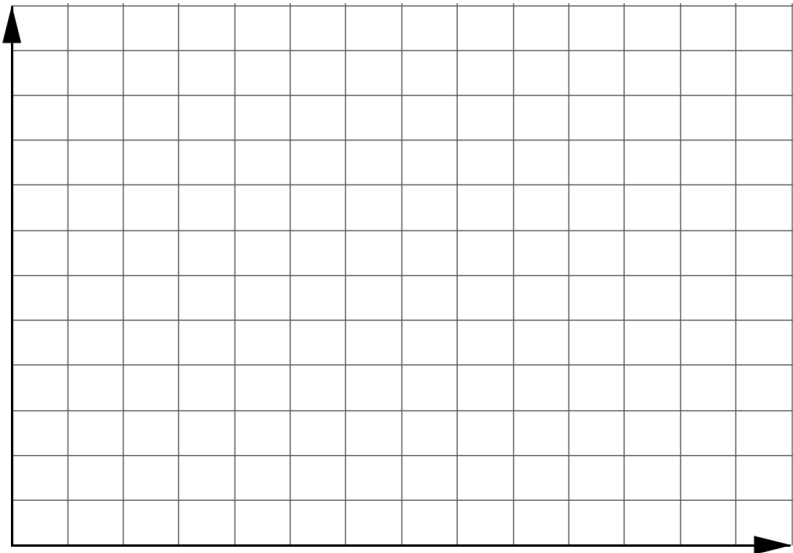
Heart Rate (bpm)	Frequency
79 – 81	2
81 – 83	7
83 – 85	10
85 – 87	7
87 – 89	4
89 – 91	2



- a. Explain what Raisa did wrong.

- b. Correct Raisa’s mistake and create a frequency table and histogram that correctly displays the data she collected.

Heart Rate (bpm)	Frequency



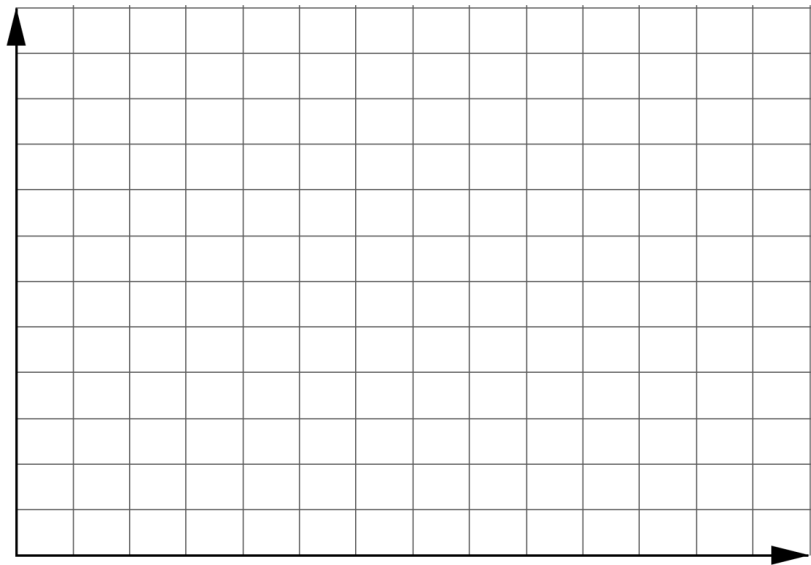
10.2.4 Your Turn!

1. Sergio noticed that a bag of chips advertised that the serving size was 15 chips and that the bag contained approximately 4 servings. He counted the number of chips in a sample of 12 bags and recorded the total number of chips below.

48, 51, 52, 55, 56, 58, 59, 60, 61, 62, 65, 67

Create a frequency table and histogram for the number of chips in a bag.

Number of Chips	Frequency



10.2.5 Wrap-Up: Reflection

1. What was the **muddiest** (most difficult or confusing) point in today's lesson on creating histograms?

